

Mock exam Introduction to Psychology and its Methods AI

This exam consists of 63 questions. Each question has an equal weight in the assessment.

1. Variability within one species is a(n) evolutionary change.

- A. unwanted by-product of
- B. essential condition for
- C. barrier to
- D. precursor of

2. When a property is recessively inherited, then

- A. it is a detrimental feature.
- B. this property is determined by sex-linked chromosomes.
- C. the property must be passed by both parents to express itself in the phenotype.
- D. only parents who have this feature can pass it to their children.

3. Which of the following options is the most direct determinant of gene expression?

- A. The moment in development
- B. The biochemical environment within a cell
- C. The biochemical environment outside a cell
- D. The total environment

4. A brother and a sister with the same parents probably have

- A. the same set of 46 chromosomes.
- B. no chromosome in common.
- C. about 50% of the chromosomes in common.
- D. the same chromosome except the sex chromosome.

5. When an individual is homozygous for a particular gene, then this individual has

- A. two recessive alleles.
- B. two dominant alleles.
- C. one recessive and one dominant allele.
- D. two identical alleles.

6. Xena the Warrior Princess was taking a nap after a big lunch. Her companion Gabrielle was attacked nearby. Xena immediately rose up and threw her weapon at the attacker, saving Gabrielle. Which of Xena's nervous system were activated during this time?

- A. Parasympathetic nervous system during the nap; Sympathetic nervous system during the attack
- B. Sympathetic nervous system during the nap; Parasympathetic nervous system during the attack
- C. Somatic nervous system during the nap; Autonomic nervous system during the attack
- D. Autonomic nervous system during the nap; Somatic nervous system during the attack

7. The two parts of the peripheral nervous system are

- A. the brain and spinal cord.
- B. the somatic and the autonomous nervous system.
- C. the sympathetic and the parasympathetic nervous system.
- D. the autonomous and sympathetic nervous system.

8. Neurotransmitters from the presynaptic neuron A have an excitatory effect on the postsynaptic neuron B as

- A. they increase the potential difference between the inside and outside of the presynaptic neuron A.
- B. they lower the potential difference between the inside and outside of the presynaptic neuron A.
- C. they increase the potential difference between the inside and outside of the postsynaptic neuron B.
- D. they lower the potential difference between the inside and outside of the postsynaptic neuron B.

9 During a brain operation, a part of the primary motor projection area is stimulated in a patient. As a result of this stimulation, the left leg of the patient moves. The most likely place of stimulation was the part of the ... lobe.

- A. left Frontal
- B. right Frontal
- C. left Parietal
- D. right Parietal

10. Some agonists cause an effect by

- A. an increase in reuptake.
- B. blocking the receptors.
- C. mimicking a neurotransmitter.
- D. a decrease of a precursor.

11. In which of the following techniques is the electrical activity of the brain recorded?

- A. fMRI
- B. PET
- C. EEG
- D. CT

12. While driving to a store, your mind wanders, and you are surprised to find that you have arrived. This experience supports the idea that:

- A. automatic processes need awareness.
- B. automatic processes do not need awareness.
- C. awareness needs automatic processes.
- D. awareness does not need automatic processes.

13. In the technique known as shadowing, a research participant repeats back:

- A. text they are reading.
- B. their internal monologue.
- C. words heard in one ear.
- D. their current conversation.

14. Frank Tong and colleagues superimposed a house onto a face and asked participants to make judgments about what they saw. What were the participants' brain responses?

- A. The brain activity changed depending on whether participants reported seeing a house or a face.
- B. The brain noticed both equally, meaning that eventually everything becomes conscious.
- C. The unconscious brain became confused and thus the participants were unable to see either the house or the face.
- D. The brain emitted large, slow-wave activity equally for both images.

15. Which of the following statements regarding the brain during sleep is NOT correct?

- A. Sleep is an altered state of consciousness.
- B. The brain shuts down during sleep.
- C. Complex thinking can occur during sleep.
- D. Conscious experience of the outside world is largely turned off during sleep.

16. An observer sees two equal trees. One tree is at a distance of 20 meters from the observer and the other tree is at a distance of 100 meters from the observer. The retinal image of the first tree will be ... the second tree.

- A. larger than the retinal image of
- B. smaller than the retinal image of
- C. equal to the retinal image of
- D. the inverse of the retinal image of

17. Which of the following statements is true?

- A. The density of cones and rods in the retina increases from the fovea to the periphery.
- B. In the fovea there are mainly rods, while the cones are mostly in the periphery.
- C. In the retina there are no cones and rods where the optic nerve exits the eye.
- D. All of the above options are correct.

18. Suppose the distance between the eyes can be increased. This would have the following effect:

- A. an increase in binocular disparity.
- B. a decrease in binocular disparity.
- C. an increase in the effect of motion parallax.
- D. a reduction in the effect of motion parallax.

19. A subject is asked how he perceives the following stimulus: -* -* -*

The subject reports that this stimulus is perceived as three pairs, each of which contains a dash and an asterisk. This subject grouped the stimuli based on

- A. similarity.
- B. proximity.
- C. continuity.
- D. closure.

20. The path that brings visual information to the parietal lobe is the

- A. What system.
- B. Where system.
- C. Why system.
- D. When system.

21. A tap with a hammer just under the knee leads to a lower leg movement, the knee reflex. This movement that occurs after a tap with a hammer is an example of a(n)

- A. Unconditioned response.
- B. Conditioned response.
- C. Unconditioned stimulus.
- D. Conditioned stimulus.

22. In classical conditioning an animal learns the relation between; In instrumental conditioning, an animal learns the relationship between

- A. CR and US; Stimulus and response
- B. CS and UR; Stimulus and response
- C. CR and UR; Response and reward
- D. CS and US; Response and reward

23. Which rat will show the fastest extinction of pushing a button? A rat trained with

- A. a Variable Ratio scheme of reinforcement.
- B. a Variable Ratio scheme of reinforcement.
- C. a Fixed Ratio scheme of reinforcement.
- D. Continuous reinforcement.

24. A child is continuously seeing the same stimulus. Eventually, the child loses any interest in the stimulus. This decrease in response represents

- A. the learning curve.
- B. the law of effect.
- C. habituation.
- D. stimulus generalization.

25. In a memory task, subjects are sometimes asked to count back during the interval between stimulus presentation and reporting. Why is this back-counting task used?

- A. This task contributes to a higher concentration.
- B. This task keeps the subject alert.
- C. This task prevents interference from other matters.
- D. This task overwrites the contents of working memory.

26. In a verbal learning study, retroactive interference proved to be a major cause of forgetting. This study showed that

- A. the probability of remembering information decreased with the amount of information previously learned.
- B. the probability of remembering information increased with the amount of information previously learned.
- C. the probability of remembering information decreased with the amount of information learned later.
- D. the probability of remembering information increased with the amount of information learned later.

27. The name of your primary school teacher is probably saved in your When you try to recall this name, you are basically engaged in an ... memory task.

- A. long-term memory; explicit
- B. long term memory; implicit

- C. random access memory; explicit
- D. random access memory; implicit

28. Which of the following findings is an indication that analogue (mental) images are not the same as perceptual images.

- A. Mental images cannot be rotated.
- B. Mental images cannot be scanned.
- C. Mental images cannot easily be reinterpreted.
- D. Mental images can easily be reinterpreted.

29. Max knows one person from New Zealand. This person is friendly and considerate. When asked what he thinks people in New Zealand are like, Max answers that he thinks they are friendly and considerate. This judgment is probably based on

- A. the representativeness of heuristics.
- B. the availability heuristics.
- C. framing effects.
- D. the confirmation bias.

30. A person must solve a very difficult problem. In order to increase the probability that he actually comes up with the solution, one needs

- A. to maximize the reward for solving the problem.
- B. to encourage the person to use analogies.
- C. to reinforce the mental set of the person.
- D. all of the above options are correct.

31. Judi is six years old and in a Binet intelligence test she solves all the items that an average nine-year-old can solve. She cannot solve the items that can be resolved by an average 10-year-old. Judi's IQ is according to Binet:

- A. 50 points.
- B. 100 points.
- C. 150 points.
- D. impossible to determine based on the above.

32. Among the group of people with an IQ lower than 70, one can distinguish between people with an IQ between 40 and 55 and people with an IQ between 55 and 70.

Which statement is correct?

- A. There are fewer people with an IQ between 40 and 55 than people with an IQ between 55 and 70.
- B. There are as many people with an IQ between 40 and 55 as people with an IQ between 55 and 70.
- C. There are more people with an IQ between 40 and 55 than people with an IQ between 55 and 70.
- D. There is no way to decide.

33. According to Charles Spearman, the intercorrelations between WAIS-R subtests are not perfect because

- A. people vary too much in their intelligence score.
- B. there is a general intelligence factor.
- C. there are specific intelligence factors.
- D. people vary too little in their intelligence score.

34. When the heritability ratio is high, behavior is predominantly determined by

- A. genetic factors.
- B. environmental factors.
- C. phenotype.
- D. differences in education.

35. Which of the following statements is correct?

- A. Science is only concerned with "visible" issues and uses the scientific method.
- B. Faith is only concerned only with "invisible" matters in a non-scientific way.
- C. a and b are correct.
- D. a and b are incorrect.

36. A good test of a hypothesis requires two conditions. Which conditions are required?

- A. 1) If [H or IC or AA] then P;
2) If [non-H or IC or AA] then very probably not P
- B. 1) If [H and IC and AA] then P;
2) If [non-H and IC and AA] then very probably not P
- C. 1) If [H or IC or AA] then P;
2) If [non-H and IC and AA] then very probably not P
- D. 1) If [H and IC and AA] then P;
2) If [non-H or IC or AA] then very probably not P

37. If there is scientific justification for a particular belief, this means that

- A. this belief is based on a controlled laboratory study.
- B. this belief is based on the correct application of the scientific method.
- C. this belief relates to "visible" matters.
- D. All the above options are correct.

38. On the basis of a hypothesis, one can formulate a prediction by On the basis of data one can formulate a hypothesis by

- A. induction; deduction
- B. deduction; induction
- C. diversion; reduction
- D. reduction; diversion

39. Which of the options below is the correct conclusion of the following conditional argument:

If P then Q

not Q

- A. P
- B. Q
- C. not P
- D. not Q

40. The variable that is manipulated in an experiment is the

- A. dependent variable.
- B. independent variable.
- C. confounding variable.
- D. control variable.

41. A researcher wants to know what the effects are of font on the reading speed of people. He performs an experiment with the ... font and the ... reading speed. All test subjects are given the task to first reading a paragraph in "Courier" font and then read a paragraph in "CG Times" font. In this experiment, font order is a

- A. dependent variable; independent variable; confounding variable
- B. independent variable; dependent variable; confounding variable
- C. dependent variable; independent variable; control variable
- D. independent variable; dependent variable; control variable

42. A researcher is interested in whether Sorbitol causes cancer in humans. He carries out a well-controlled laboratory experiment in which he uses two identical groups of rats.

The experimental group receives a large amount of Sorbitol every day, while the control group receives a placebo every day. The results show that there are significantly more cases of cancer in the experimental group than in the control group. The researcher's conclusion is that intake of Sorbitol causes cancer. This study has a

- A. high internal validity and high external validity.
- B. high internal validity and low external validity.
- C. low internal validity and high external validity.
- D. low internal validity and low external validity.

43. A good operational definition of a variable is in particular important for the

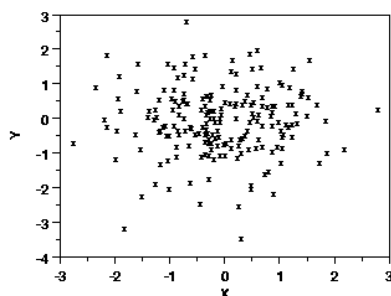
- A. construct validity.
- B. internal validity.
- C. predictive validity.
- D. external validity.

44. In which of the following options is the internal validity probably the greatest?

- A. descriptive study
- B. N = 1 study
- C. correlational study
- D. experimental study

45. A ratio scale has the following property:

- A. differences.
- B. equal intervals.
- C. absolute zero.
- D. All the above options are correct.



46. What is the correlation between X and Y in the scatter diagram below?

- A. about -1
- B. about 0
- C. about 1
- D. The correlation can only be determined when there is sufficient variability

47. Within the cross-lagged-panel procedure, it can be concluded that a variable A affects a variable B when

- A. the correlation between variable A measured at time 1 and variable B measured at time 2 is larger than the correlation between variable A measured at time 1 and variable B measured at time 1.
- B. the correlation between variable A measured at time 1 and variable B measured at time 1 is greater than the correlation between variable A measured at time 1 and variable B measured at time 2.
- C. the correlation between variable A measured at time 1 and variable B measured at time 1 is greater than the correlation between variable A measured at time 2 and variable B measured at time 2.
- D. the correlation between variable A measured at time 2 and variable B measured at time 2 is greater than the correlation between variable A measured at time 1 and variable B measured at time 1.

48. An exploratory study yields the following results:

Test subject	Weight (kg)	Height (cm)
1	70	170
2	75	175
3	80	190
4	65	165
5	70	170
6	90	190

On the basis of the above results one can state that (hint: draw the results in a scatter diagram)

- A. there is a positive correlation between weight and height.
- B. there is a negative correlation between weight and height.
- C. there is a correlation of about 0 between weight and length.
- D. the correlation cannot be determined on the basis of too few subjects.

49. In between-subjects design there is no chance on ... but on ...

- A. carry-over effects; group differences
- B. group differences; carry-over effects
- C. contamination; group differences
- D. group differences; contamination

test person	age	number of words remembered
1.	5	3
2.	10	5
3.	15	7
4.	20	8
5.	30	8
6.	40	8
7.	50	8
8.	60	5
9.	65	3
10.	80	3

50. To determine whether there is a correlation between age and memory capacity (number of words remembered in memory task), 10 people were measured. This yielded the following data:

(hint: draw the results in a scatter diagram)

- A. There is no linear correlation between age and number of remembered words.
- B. There is a negative correlation between age and number of remembered words.
- C. There is a positive correlation between age and number of remembered words.
- D. The correlation cannot be determined.

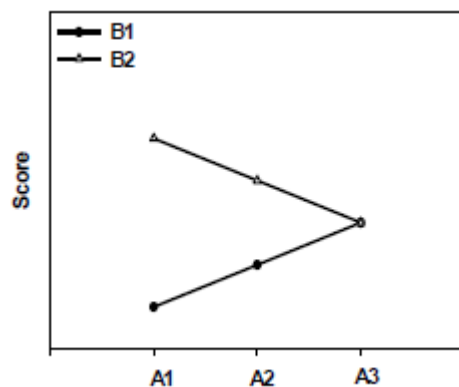
51. An independent variable with 4 levels is within-subjects manipulated. The researcher uses a balanced Latin Square design. How many subjects should at least participate in this study?

- A. 4
- B. 8
- C. 16
- D. 24

52. An experiment uses a 5 x 2 factorial between-subjects design. How many different groups of subjects are participating in the experiment?

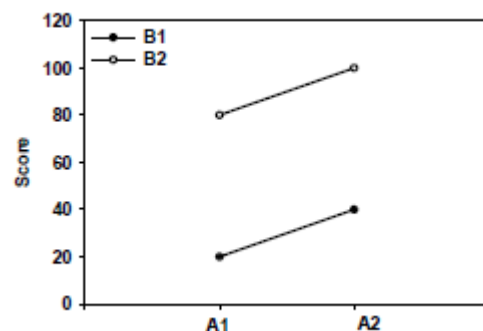
- A. 1
- B. 5
- C. 7
- D. 10

53. The graph below shows



- A. a main effect of A; a main effect of B; an interaction between A and B.
- B. a main effect of A; no effect of B; an interaction between A and B.
- C. no effect of A; a main effect of B; an interaction between A and B.
- D. a main effect of A; a main effect of B; no interaction between A and B.

54. The graph below shows



- A. a main effect of A; a main effect of B; an interaction between A and B.
- B. a main effect of A; no effect of B; an interaction between A and B.
- C. no effect of A; a main effect of B; an interaction between A and B.
- D. a main effect of A; a main effect of B; no interaction between A and B.

55. In an experiment with a 2 x 2 factorial design, the following results are obtained:

		Independent variable A	
		A1	A2
Independent variable B	B1	100	140
	B2	120	160

In this experiment there is

- A. a main effect of A; a main effect of B; an interaction between A and B.
- B. a main effect of A; no effect of B; an interaction between A and B.
- C. no effect of A; a main effect of B; an interaction between A and B.
- D. a main effect of A; a main effect of B; no interaction between A and B.

56. A disadvantage of a quasi-experimental design compared to a factorial design in which the independent variables are manipulated is

- A. a lower internal validity.
- B. a lower reliability.
- C. a lower external validity.
- D. a lower construct validity.

57. In an N = 1 study an AB design is used. The results show that there are clear differences between episode A and episode B on the dependent variable. The conclusion based on this research can be that

- A. the treatment has an effect.
- B. the treatment has no effect.
- C. there is confounding.
- D. All the above options are possible.

58. A teacher was interested in the development of problem solving. He conducted an experiment with 100 five-year-olds and 100 seven-year-olds. Each participant had to solve a puzzle problem. Each participant was allowed to take as much time as necessary. 99% of the seven-year-olds and 98% of the five-year-olds were able to solve the puzzle. Since the difference between the two groups was not statistically significant, the teacher concluded that there is no difference between five- and seven-year-olds in the ability to solve puzzle problems. The above results are probably caused by a

- A. regression-to-average effect.
- B. ceiling effect.
- C. floor effect.
- D. confounding variable.

59. When using a non-equivalent control group, it is important

- A. draw both groups randomly from the population.
- B. match both groups with respect to possible contaminating variables.
- C. assign the subjects randomly to the groups.
- D. All options are correct.

60. Which of the following variables is not a person variable?

- A. age
- B. sex
- C. memory skills
- D. calculation instructions

1. b	16. a	31. c	46. b
2. c	17. c	32. a	47. a
3. b	18. a	33. c	48. a
4. c	19. b	34. a	49. a
5. d	20. b	35. d	50. a
6. a	21. a	36. b	51. a
7. b	22. d	37. b	52. d
8. d	23. d	38. b	53. c
9. b	24. c	39. c	54. d
10. c	25. d	40. b	55. d
11. c	26. c	41. b	56. a
12. b	27. a	42. b	57. d
13. c	28. c	43. a	58. b
14. a	29. a	44. d	59. b
15. b	30. b	45. d	60. d